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FEDERAL BOARD OF INTERMEDIATE AND SECONDARY EDUCATION ISLAMABAD

COMPUTER SCIENCE
HSSC–I
SECTION – A (Marks 13)

Time allowed: 20 Minutes

Section – A is compulsory. All parts of this section are to be answered on this page and handed over to the Centre Superintendent. Deleting/overwriting is not allowed.

Do not use lead pencil.

حصّہ اوّل لازمی ہے۔ اس کے جوابات اسی صفحہ پر دے کر ناظم مرکز کے حوالے کریں۔ کاٹ کر دوبارہ لکھنے کی اجازت نہیں ہے۔ لیڈ پنسل کا استعمال ممنوع ہے۔

Version No.				
3	2	0	7	1

ROLL NUMBER						

Answer Sheet No. ___________

Invigilator Sign. ___________

ہر سوال کے سامنے دیے گئے، کریکولم کے مطابق درست دائرہ کو پر کریں۔

Fill the relevant bubble against each question according to curriculum:

Candidate Sign. ___________

	Question	A	B	C	D	A	B	C	D
1.	Which gate produces a LOW output when **at least** one input is HIGH?	AND gate	NAND gate	NOR gate	OR gate	○	○	○	○
2.	What is an efficient algorithm for finding a specific book in a large library?	Asking other visitors for help	Using the catalog system to find its location	Randomly pulling out books from the shelves	Browsing without a plan	○	○	○	○
3.	Which of the following represents a reliable source of information?	A personal blog with no evidence	A social media post without citation	A peer-reviewed academic journal	A random website with advertisements	○	○	○	○
4.	**K-means clustering** is an example of the following statistical modelling technique:	Supervised learning	Semi-supervised learning	Reinforcement learning	Unsupervised learning	○	○	○	○
5.	The key advantage of using a private cloud is:	Higher security and control over data	Lower cost compared to public cloud	Accessible to the public	Reduced need for IT infrastructure	○	○	○	○
6.	Consider the list, **A= [10, 20, 30, 40]**, what will the statement **print (len(A))** return?	0	4	40	Error in statement	○	○	○	○
7.	What does decomposition mean in computational thinking?	Combining smaller problems into a larger solution	Ignoring the problem until it resolves itself	Writing code without any structure	Breaking a problem into smaller, more manageable parts	○	○	○	○
8.	Which type of feasibility study in SDLC assesses whether the proposed system can be completed within a **given timeframe**?	Legal feasibility	Schedule feasibility	Economic feasibility	Operational feasibility	○	○	○	○
9.	Which of the following is an example of a typical survey question?	Solve *5x7=?*	What is the capital of China?	How many hours do you spend on the internet daily?	What is the boiling point of water?	○	○	○	○
10.	Which of the following can be classified as a low-fidelity prototype?	A wireframe sketch on paper	A fully functional mobile app	A high-resolution mock up with animations	A completed website with backend functionality	○	○	○	○
11.	Which of the following is a characteristic of qualitative interviews?	They have only one-word answers	They focus on open-ended questions	They only collect numerical data	They follow a strict set of yes/no questions	○	○	○	○
12.	What will be the result of $x = 5 \& 3$ Python statement?	1	3	5	7	○	○	○	○
13.	Which of the following is an example of a product?	Online banking	Car repair	Mobile phone	Medical consultation	○	○	○	○

40

COMPUTER SCIENCE HSSC–I

Time allowed: 2:40 Hours **Total Marks Sections B and C: 62**

SECTION – B (Marks 42)

Q. 2 Answer the following parts briefly. **(14 x 3 = 42)**

<table>
<tr><td>(i)</td><td>Assume the variables x, y and z where x=4, y=8, and z=2. What value will be stored in result1, result2 and result3 after executing following Python statements?
a) result1 = x + y
b) result2 = z ** 3
c) result3 = y/x</td><td>03</td><td>OR</td><td>Trace the following pseudocode and complete the table below, which tracks the values of the variables at each iteration of the loop:
SET sum=0
FOR i FROM 1 TO 3 DO
  sum = sum + i
END FOR
PRINT sum
<table><tr><td>Iteration no.</td><td>Value of i</td><td>Value of sum</td><td>Print</td></tr><tr><td>1</td><td></td><td></td><td></td></tr><tr><td>2</td><td></td><td></td><td></td></tr><tr><td>3</td><td></td><td></td><td></td></tr></table></td><td>03</td></tr>
<tr><td>(ii)</td><td>What is a use case for statistical modelling? Give an example.</td><td>2+1</td><td>OR</td><td>Write down three benefits of integration of Blockchain and IoT.</td><td>03</td></tr>
<tr><td>(iii)</td><td>How does Agile model differ from Waterfall model? Justify with at least three reasons.</td><td>03</td><td>OR</td><td>Differentiate between Parameter and Statistics with daily life examples.</td><td>1.5
+
1.5</td></tr>
<tr><td>(iv)</td><td>Why is successful product development important? Give three reasons.</td><td>03</td><td>OR</td><td>Write down three differences between analog and digital signals.</td><td>03</td></tr>
<tr><td>(v)</td><td>Determine whether these variable names are valid or invalid. Provide reasons for any invalid variable names.
a) range b) total marks c) counter</td><td>03</td><td>OR</td><td>How do computer systems protect against cyber-attacks? Provide three reasons.</td><td>03</td></tr>
<tr><td>(vi)</td><td>Briefly explain three guidelines regarding the safe and responsible use of information sources.</td><td>03</td><td>OR</td><td>What are embedded systems? Enlist two of their applications.</td><td>1+2</td></tr>
<tr><td>(vii)</td><td>Which algorithm is more efficient: insertion sort or bubble sort? Justify your choice with two reasons.</td><td>1+2</td><td>OR</td><td>Correct any three errors in the following Python code:
number = 0;
if number > 0:
  print(“Positive number)
eliff number = 0:
  print(“Zero”)
else:
print(“Negative number”)</td><td>03</td></tr>
<tr><td>(viii)</td><td>Evaluate the following expression in Python using the proper order of operations:
3**2-8%4+10/5*3</td><td>03</td><td>OR</td><td>Compare Star and Ring network topologies with respect to design, scalability and reliability.</td><td>03</td></tr>
<tr><td>(ix)</td><td>Write down any three features of infographics for data representation.</td><td>03</td><td>OR</td><td>Write down any three key steps in developing the Minimum Viable Product.</td><td>03</td></tr>
<tr><td>(x)</td><td>Write a Python program that asks the user to input name and age also displays the entered name and age.</td><td>03</td><td>OR</td><td>Draw truth table for the Boolean function.
$F=XYZ+X\overline{Y}Z$</td><td>03</td></tr>
<tr><td>(xi)</td><td>Write down three features of the Survey for data collection.</td><td>03</td><td>OR</td><td>What is meant by data searches? Illustrate with an example.</td><td>2+1</td></tr>
<tr><td>(xii)</td><td>How does vertical scalability differ from horizontal scalability in cloud computing? Illustrate with three differences.</td><td>03</td><td>OR</td><td>What will be generated by the following Python code?
list1=[34, “abc”, 40, 89]
list1[2]=67
list1.append(“xyz”)
print(list1)</td><td>03</td></tr>
<tr><td>(xiii)</td><td>What is the difference between a population and a sample in statistics? Illustrate with example.</td><td>03</td><td>OR</td><td>Write Python statements considering turtle graphics to perform the following actions:
a) Rotate the turtle 45 degree to the right.
b) Move the turtle forward by 100 pixels.
c) Change the pen colour to red.</td><td>03</td></tr>
<tr><td>(xiv)</td><td>What is the deployment phase in SDLC? How does the direct deployment method differ from the phased deployment method?</td><td>1+2</td><td>OR</td><td>Compare supervised and unsupervised learning with daily life examples.</td><td>1.5
+
1.5</td></tr>
</table>

SECTION – C (Marks 20)

Attempt the following questions. **(4 x 5 = 20)**

<table>
<tr><td>Q.3</td><td>What is encryption? Differentiate between symmetric and asymmetric encryption in terms of speed, complexity, security and key structure.</td><td>1+4</td><td>OR</td><td>What is data collection? Compare primary and secondary data collection methods based on the originality, cost, time required and reliability.</td><td>1+4</td></tr>
<tr><td>Q.4</td><td>Using a 3-variable Karnaugh Map, simplify the following Boolean Function F. Also represent the simplified Boolean expression with a logic circuit diagram.
$F=A\overline{B}\overline{C}+\overline{A}\overline{B}C+A\overline{B}C+\overline{A}BC+ABC+\overline{A}B\overline{C}$</td><td>05</td><td>OR</td><td>Write a Python program that reads three numbers and prints their product by using function.</td><td>05</td></tr>
<tr><td>Q.5</td><td>Use the Binary Search algorithm to find the number, 75 from the given set of numbers:
<table><tr><td>9</td><td>10</td><td>26</td><td>34</td><td>41</td><td>58</td><td>67</td><td>75</td><td>83</td></tr></table></td><td>05</td><td>OR</td><td>Write a Python program that takes a number and prints its factorial using for loop.</td><td>05</td></tr>
<tr><td>Q.6</td><td>What does Artificial Intelligence mean? Describe any four positive impacts of AI systems.</td><td>1+4</td><td>OR</td><td>What are assistive technologies? Provide four examples of their everyday uses.</td><td>1+4</td></tr>
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